ISA 444: Business Forecasting

23 - Seasonal ARIMA Models

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Spring 2022

Outline

1 **Preface**

2 Seasonal ARIMA Models

Recap of What we Have Covered Last Week

Main Learning Outcomes

- Describe AIC, AICc, and BIC and how they are used to measure model fit.
- Describe the algorithm used within the auto.arima() function to fit an ARIMA model.
- Describe the results of the auto.arima() function.
- Applying the rolling origin concept using R (will not be on the exam)
- Forecasting many time-series using a nested structure (will also not be on the exam)

A Comment on the map() Function

```
btc_df = readRDS("btc_histDay.rds")

btc_df %>% ggplot(aes(x = date, y = close)) + geom_line()

nested_df = btc_df %>%
  # keeping only relevant cols
  select(date, close) %>%
  rsample::rolling_origin(initial = which(btc_df$date == "2020-12-31"),
                  assess =  1, # predict 1-day ahead
                  cumulative = TRUE) # each training sample grows by 1 day

nested_df %<>% mutate(
  data = map(.x = splits, .f = rsample::analysis),
  # extracting both the train dates and their associated closing price
  trainDates = map(.x = data, .f = `[[`, 'date' ),
  trainClose = map(.x = data, .f = extract2, 'close'),
  # extracting the target data
  target_data = map(.x = splits, .f = rsample::assessment),
  target_date = map(.x = target_data, .f = extract2, 'date'),
  target_close = map_dbl(.x = target_data, .f= extract2, 'close')
)

head(nested_df)

results_df = nested_df %>%
  select(-c(data, trainDates, target_data, target_date))

results_df %<>%
  mutate(autoArima = map(.x = trainClose, .f = auto.arima) %>%
           map(.f = forecast, h = 1) %>%
```

Learning Outcomes for Today's Class

Main Learning Outcomes

- **Describe a seasonal ARIMA model and explain how it applies to a seasonal time series.**
- **Recognize when to fit a seasonal ARIMA model.**

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Live Coding: Example 1

```
pacman::p_load(astsa)
birthData = birth # also from the astsa package
```

In-Class Activity

Use the data "23 - Netflix_growth_pct_2000.csv".

Fit an ARIMA model using the auto.arima() function. Describe the model that is fit, evaluate the model residuals. Please answer the following questions.

1. Based on the summary function (applied to your model object), please explain what are the AR lag coefficients included in the model?
2. What does the seasonal ARIMA, D, parameter value indicate?
3. What did we learn from the non-seasonal ARIMA parameters?
4. In terms of the predictive performance, do we have a reasonable model?
5. In terms of the assumptions behind the ARIMA model, do we have a reasonable model?

Things to Do to Prepare for Exam 3 (April 22–25, 2022)

- Go through the slides, examples and make sure you have a good understanding of what we have covered. There will be a larger focus on interpreting the outputs from R.
- Given that several of you expressed the need for more time on the exam, the exam will be virtual and you will have 120 minutes to complete it. **Note that the exam is still designed for 80 minutes.**
- **Proctorio** will be required and I will be **recording your screen, audio, video, and web traffic**. You will need to be using Google Chrome with the Proctorio add-in installed.

Exam 3 Information: General Guidelines

- Individual assignment.
- This is a timed assignment. The exam is designed to be completed in 80 minutes – you have 120 minutes so that time is not an issue at all.
- You have to complete this assignment in one sitting.
- Open-notes (you can open other tabs, make sure you do not close the one for the "quiz").
- The quiz is made of 25 multiple choice questions, each worth 4 points.
- Proctorio is required (see previous slide).

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